



lecture 21

let's say the following transfer function is given

$$G(s) = \frac{10}{s^2 + 2s + 10}$$

compute i) rise time

ii) peak time

iii) % overshoot

iv) peak value

v) settling time

vi) Draw the step response

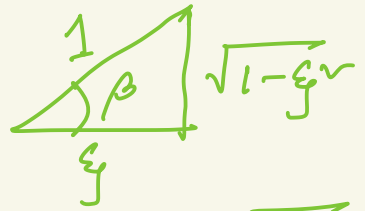
$$\omega_n = \sqrt{10} \text{ Hz} \quad \zeta = \frac{1}{\sqrt{10}}$$

i) Rise time:

$$t_r = \frac{\pi - \beta}{\omega_d}$$

$$= \frac{3.1414 - 1.249}{\omega_n \sqrt{1 - \xi^2}}$$

$$= () \text{ sec}$$



$$\tan \beta = \frac{\sqrt{1 - \xi^2}}{\xi}$$

$$\beta = 1.249 \text{ rad}$$

$$\text{iii) } \%M_p = e^{-\xi\pi/\sqrt{1-\xi^2}} \times 100\%$$

$$= 35.12\%$$

$$\text{ii) } t_p = \frac{\pi}{\omega_n \sqrt{1 - \xi^2}}$$

$$\text{iv) } y(t_p) = 1 + 0.3512 = 1.3512$$

$$\text{v) } t_s = \frac{4.6}{\xi \omega_n} =$$

vij) The response was drawn in the class.

Let's say the following transfer function is given.

$$G(s) = \frac{50}{s^2 + 6s + 25}$$

Find t_s , t_p , y_p , M_p .

t_s , t_p and M_p can be found easily by following the process shown before. For finding y_p slight modification in the transfer function is needed.

$$G(s) = \frac{50}{s^2 + 6s + 25}$$

$$= 2 \cdot \frac{25}{s^2 + 6s + 25}$$

$$= K \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}$$

$$K = 2 \quad (\text{DC gain})$$

$$t_s =$$

$$t_p =$$

$$M_p = 0.095 = 9.5\%$$

$$y_p = 2(0.095) + 2$$

$$= 2.19$$

